

From Social Media to Mental Well-being: Modelling the Spread of Emotion in Online Networks

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BACKGROUND CONTEXT

Social media platforms have fundamentally transformed how information spreads through populations, generating new challenges for understanding their impact on mental health. Anchored in the frameworks of emotional contagion and infodemiology (the spread of information as it relates to public health), this research investigates how affective information diffuses through online social networks and its implications for individuals with depression.

This interdisciplinary research situated within the theme of ‘**Health and Wellbeing**’ involved both computational and qualitative research methodologies to identify, interpret, measure and model information as affective and diffusive within online social networks. As social media continues to evolve and data access becomes increasingly restricted, this research provides timely evidence for the urgent need to prioritise mental health considerations in digital platform design and public health discourse.

ACADEMIC RESEARCH OUTPUTS

[2]

PUBLISHED
MANUSCRIPTS

[5]

LOCAL AND
INTERNATIONAL
CONFERENCES

[5+]

MEDIA
RELEASES

[1]

ARTICLES
UNDER REVIEW

[2]

MANUSCRIPTS
IN PREPARATION

METHODOLOGIES

[1]

Empirical content analysis of Reddit data

Collected 11 950 Reddit posts and analysed responses based on general engagement, temporal attributes, as well as demographic and geographic attributes.

[2]

Topic modelling and thematic analysis of Reddit corpus data

Combined computational natural language processing and qualitative thematic analysis applied to r/depression subreddit data to identify four major themes and 23 sub-themes based on the nature of the Reddit posts.

[3]

Construction of a depression-specific sentiment lexicon

Used subreddit data to construct a depression-specific sentiment lexicon that better assigns polarity labels (**positive**, **neutral** and **negative**) to depressotypic texts that rivals open-source general-purpose lexica.

[4]

Agent-based simulation of sentiment spread within depression network on Twitter/X

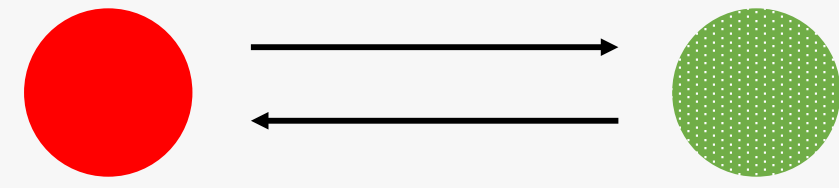
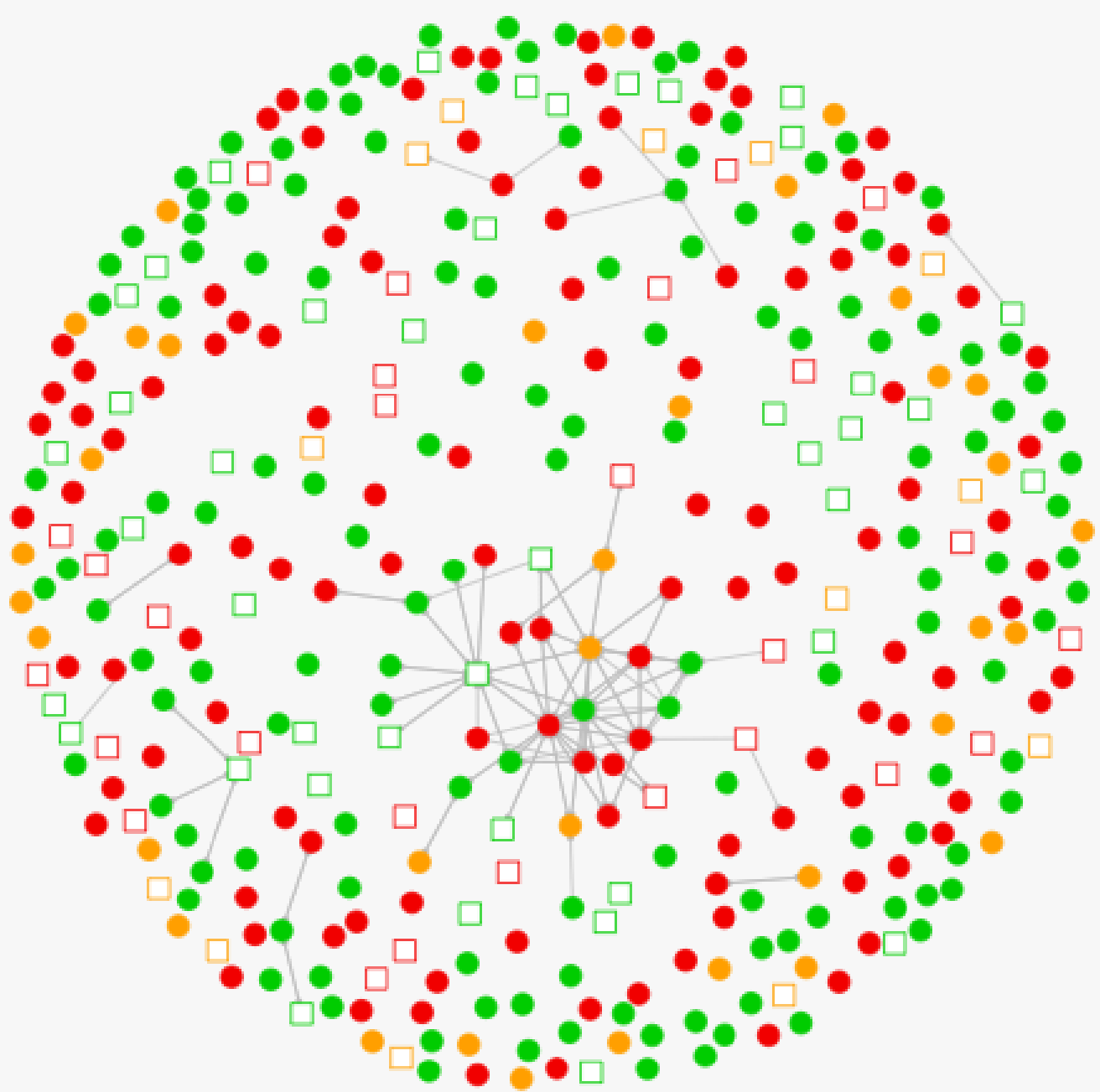
Simulated a representative social network of 379 Twitter users sharing their experiences with depression, as informed by themes and markers of depression identified from subreddit data.

[5]

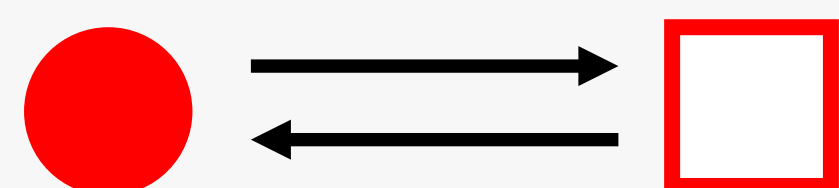
Intervention strategies for online communication on depression network on Twitter/X

Applied individual- and group-level interventions to simulated network to improve well-being (positive states of sentiment shared online) and draw connections between tested scenarios and real-world applications.

KEY FINDINGS AND CONTRIBUTIONS



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Increased posting activity on certain days and at certain times

Reddit activity saw an increase in submissions to r/depression on public holidays and particular days of significance (Mothers’ Day and birthdays). Increased activity seen, on average, between 10 PM and 5 AM SAST. Most posts were generated from individuals in countries with higher prevalence of depression and typically from younger individuals (although posts ranged from users aged from 10 to 75 years old) based on data collected from April to June 2023.

Sentiment is an appropriate measure of emotion (and thus emotion spread)

Existing sentiment tools are trained on domain-independent contexts. A lack of linguistic resources exist that can account for depression-specific contexts. The construction of a novel sentiment lexicon for depression-specific texts showed improvement on the task of sentiment classification of texts compared to VADER, LIWC, MPQA, and NRC.

Emotional contagion exists in the form of emotional reinforcement

An observed phenomenon from the research was the presence of **emotional reinforcement** rather than traditional emotional contagion. In other words, rather than emotions spreading from person to person, two individuals with some social connection will experience intensified emotions when both individuals share the same diagnosis and emotion at the same time, for longer durations and more frequently compared to any other combinations [see left figure]. Individuals diagnosed with depression and in negative states of emotion are particularly vulnerable to this effect.

IMPACT BEYOND ACADEMIA

For individuals on social media

Awareness surrounding digital health interventions can and has led internet users to conduct their own research on mental health problems online. This manifests as individuals in online mental health communities seeking out advice and guidance from others’ experiences and stories, using apps related to well-being and using digital tools to connect with health practitioners. Platform users also stand to benefit from **applying their agency to curate a social media experience** through interventions that improve their well-being. This is especially true for individuals with mental health disorders who are most susceptible to emotional contagion.



For social media platform developers

Platform algorithm developers and designers also have a responsibility towards fostering a healthier environment for their users. Platform features that allow users to curate their experiences rather than be beholden to the throes of “the algorithm” are being investigated. News headlines surrounding platforms’ contribution to social media addiction and mental harm in adolescents supports the notion that platforms have a direct responsibility to its users to ensure safe digital experiences as far as possible. Proposed interventions include parental controls, time and age restrictions, night curfews and influence-free media, **all of which are currently being investigated by the major platforms**. Recommendations from this research would include advanced blocking/muting features, and increased content moderation efforts.



For mental healthcare practitioners and researchers

Mental health practitioners may benefit from the analysis of the textual data generated by individuals’ experiences with depression. The insights from this research already informs on more current experiences with depression compared to what is currently captured in available diagnostic tools. A mental health practitioner informed by social media data may also gain a better understanding of their patients or to reach people in online spaces where there is reduced stigma surrounding mental healthcare. It is also the **intention to develop the depression-specific lexicon into a functional depression inventory for patient self-evaluation** in collaboration with mental healthcare practitioners.



For governments, policymakers and society

The UN Sustainable Development Goals only make subsidiary provision for mental health and well-being (under Target 3.4, related to non-communicable diseases). Such calls to action require a greater focus on mental health for the growing global health crisis that it is. Governments in Australia, the EU and the UK, Korea, Malaysia, Indonesia, Brazil and some others have taken to regulating and/or imposing bans of social media use for minors. Comments have been provided to the **UK government for influence-free requirements** (no ads, sponsored content or content from non-connections).



FUTURE PATHWAYS TO IMPACT

Collaboration with international researchers

Opportunities to collaborate with researchers in the space are the immediate next step in this research. This involves attending conferences where this work is most applicable (such as the ACM Computer-Human Interaction conference, the International AAAI Conference on Web and Social Media, the Empirical Methods in Natural Language Processing conference and conference of the International Society for Computational Social Science). This is the best opportunity to gain academic exposure with prominent researchers in the field.

Media coverage of online mental health

The completion and public availability of the research has allowed for media engagement around the topic of social media and mental health. Highlights of the work have been shared to publishing and media establishments. The work has been shared (or has been scheduled to appear) in the local newspaper (Eikestad News), national publications (IOL News), radio (Smile FM) and television (eNCA NewsLink). [Scan for available media releases].



Development of a self-evaluation depression inventory

Depression inventories or depression scales have been developed for and are applied by clinicians, researchers and patients themselves. They are used as a means of diagnosis and screening for clinicians, for planning and monitoring treatment management strategies, for assessing the severity of the symptoms and as an accessible resource for self-evaluation and awareness. Given that stigma remains a prominent reason for individuals to not seek assistance, the development of a depression inventory using the depression-specific lexicon will provide online users with a tool to evaluate the extent to which it is necessary to seek assistance. This requires collaboration with mental healthcare practitioners, psychologists and social media experts on its development, testing and evaluation, monitoring and deployment.

Funding for developing online wellness tools

A long-term objective is to obtain funding for the development of tools to improve the well-being of individuals consuming content online. Currently supervised students are investigating ways of measuring emotions from online textual data and publicly available happiness indices. Continued supervision within this domain will lead to increased knowledge that can be applied to the development of well-being indicators. This would be in the form of efficient web browser extensions that help to identify and visually highlight or censor potentially harmful content. The funding would go to supporting doctoral and postdoctoral researchers with technical expertise, the development of the tool, maintenance of the tool as new terminology and slang develops, and expansion into other languages. The primary focus would be for low-resource languages and African languages. This will also require development of new linguistic tools, data and resources that are currently inaccessible or non-existent.

SIGNIFICANCE OF RESEARCH

Comment from manuscript reviewer (published in International Journal of Health, Wellness, and Society)

*“The article addresses a **highly relevant and timely topic**, exploring the lived experiences of individuals with depression through online mental health communities. Its thematic focus is **significant, original, and fills a critical gap in current psychiatric and sociolinguistic research** by prioritizing patient narratives over purely clinical viewpoints. The **empirical grounding is robust**, supported by a **large and ethically collected dataset from Reddit**, and analyzed through a **rigorous combination of computational topic modeling and qualitative thematic analysis**. The data collection and analytical methods are both appropriate and sufficient to address the research questions effectively. The study engages comprehensively with existing literature, demonstrating awareness of theoretical, clinical, and methodological frameworks. Furthermore, it **relates coherently to real-world applications, offering valuable insights for diagnostic refinement, therapeutic communication, and digital mental health practice**. Overall, the article is methodologically rigorous, contextually grounded, and intellectually compelling, making a **meaningful contribution to both academic scholarship and practical mental health discourse**.”*



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[Link to publications]